

Modifying the Affective Behavior of Preschoolers with Autism Using *In-Vivo* or Video Modeling and Reinforcement Contingencies

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The purpose of this study was to modify the affective behavior of three preschoolers with autism in home settings and in the context of play activities, and to compare the effects of video modeling to the effects of *in-vivo* modeling in teaching these children contextually appropriate affective responses. A multiple-baseline design across subjects, with a return to baseline condition, was used to assess the effects of treatment that consisted of reinforcement, video modeling, *in-vivo* modeling, and prompting. During training trials, reinforcement in the form of verbal praise and tokens was delivered contingent upon appropriate affective responding. Error correction procedures differed for each treatment condition. In the *in-vivo* modeling condition, the therapist used modeling and verbal prompting. In the video modeling condition, video segments of a peer modeling the correct response and verbal prompting by the therapist were used as corrective procedures. Participants received treatment in three categories of affective behavior – sympathy, appreciation, and disapproval – and were presented with a total of 140 different scenarios. The study demonstrated that both treatments – video modeling and *in-vivo* modeling – systematically increased appropriate affective responding in all response categories for the three participants. Additionally, treatment effects generalized across responses to untrained scenarios, the child's mother, new therapists, and time.

KEY WORDS: Autism; affective behavior; *in-vivo* modeling; observational learning; video modeling.

INTRODUCTION

Discriminating social cues and modifying one's affective behavior, accordingly, are both fundamentally important for human social interactions. Affective interchanges are particularly important for the

mother-child relationship during the early stages of life, and failure to use them appropriately may result in impaired communication and poor social development (Feldman, Philippot & Custrini, 1991). Deficits in discriminating and adjusting affective behavior according to social demands is a core deficit of autism spectrum disorders. Specifically, children with autism present severe deficits in displaying appropriate affect, which ultimately impedes their overall social development (McGee, Feldman, & Chernin, 1991). Those deficits include affective displays that are limited in range, seemingly unrelated to social context, or odd in appearance (American Psychiatric Association, 1994; Schreibman & Charlop-Christy, 1998). Deficits in the social-affective domain may be

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attributed to the restricted social-cognitive understanding associated with autism, which may also account for difficulties in joint attention, non-verbal communication, the production of referential language, and various other social and communication skills. The common denominator of those skills is the ability to coordinate one's attention between a social partner and various relevant environmental stimuli (Carpenter, Pennington, & Rogers, 2002). Furthermore, the development of social-communicative skills presumes a basic understanding of other people's thoughts and intentions (Bauminger, 2002). Researchers who consider lack of a theory of mind to be one of the primary deficits of children with autism, suggest that teaching children with autism to understand other people's thoughts and intentions, or developing a theory of mind, would be crucial in producing widespread, qualitative changes in social behavior and would promote important skills, such as initiations of social interactions, rather than enhance specific social and communication skills, under specific conditions (e.g. Baron-Cohen, Tager-Flusberg, & Cohen, 1993; Hadwin, Baron-Cohen, Howlin, & Hill, 1997; Howlin, Baron-Cohen, & Hadwin, 1999). Even though it is suggested that teaching skills that promote the development of a theory of mind, is of critical importance for the amelioration of social-affective deficits in children with autism, and more efficient than teaching specific social skills, the relationship between understanding the thoughts and intentions of others and the ability to express one's affect in a reciprocal manner, or other types of affective behavior, has not yet been investigated empirically. In fact, empirical investigations, so far, have failed to find a correlation between conversational skills and skills associated with the development of a theory of mind (Chin & Bernard-Opitz, 2000; Hadwin *et al.*, 1997).

A large number of studies have addressed the development of social skills in people with disabilities, yet studies that aim to modify their social affective behavior are scarce (e.g. Acker, Acker, & Pearson, 1973; Cooke & Apolloni, 1976; Field & Walden, 1982; Odom, Hoyson, Jamieson, & Strain, 1985), and only one study specifically addressed teaching appropriate affective responding to adolescents with autism (Gena, Krantz, McClannahan, & Poulson, 1996). Despite the acknowledgment that children with autism present severe affective deficits and the fact that early intervention is crucial for the treatment of autism, there are no studies addressing the acquisition of appropriate affect in young children with autism.

Various procedures have been identified that reliably produce positive changes in the social behavior of children with disabilities. Odom *et al.*, (1985) used peer imitation procedures to enhance the social skills of handicapped, socially withdrawn preschoolers. Typical peers were used as "confederates" who made social initiations, which helped to increase the number of positive social interactions between them and their handicapped peers. Even though the handicapped preschoolers accomplished an increase in positive interactions, those interactions did not generalize across settings. In another study in which instruction, modeling, and contingent social praise were used, children with learning disabilities learned to display positive social-emotional responses, such as smiling, sharing, complimenting, and positive physical contact. Treatment outcomes maintained during follow-up and generalized to untrained peers who interacted with the subjects (Cooke & Apolloni, 1976).

In a more recent study, Charlop-Christy, Le, and Freeman (2000) demonstrated that video modeling was even more effective than *in-vivo* modeling in teaching children with autism various tasks, including social skills (e.g. spontaneous greetings). Video modeling, as *in-vivo* modeling, promotes learning through observation. Observational learning is said to occur when an observer matches responses performed by a model and requires complex discriminations as well as attending to multiple cues, such as the modeled response, the context, and the consequences of the modeled behavior (Goldstein & Brown, 1989). Thus, for children with autism who have difficulties in developing imitative repertoires (Rogers, 1999), as well as attending to multiple cues, learning through observation may be a challenging, yet a very important medium for learning. Several studies have demonstrated that modeling may enhance the development of simple discriminations (e.g., Young, Krantz, McClannahan, & Poulson, 1994), as well as complex responses, such as independent play (e.g., Tryon & Keane, 1986) or gestural communication skills (Buffington, Krantz, McClannahan, & Poulson, 1998) in children with autism, even for those with poor imitative repertoires. Learning through observation of peers has also been effective in teaching children with autism various skills, including complex social skills, such as social interactions with peers (Garfinkle & Schwartz, 2002), and was found to promote generalization to a greater extent than more traditional teaching procedures, such as discrete-trial teaching and modeling provided by adults (Charlop,

Schreibman, & Tryon, 1983; Ihrig & Wolckik, 1988). More recently, observational learning with the use of video models has also been used with children with disabilities, in general, and with autism in particular (e.g., Dorwick, 1991, 1999; Haring, Kennedy, Adams, & Pitts-Conway, 1987). Charlop-Christy and her colleagues have developed video-modeling procedures that are most efficient in teaching language, social, and recreational skills as well as promoting generalization of such skills in children with autism (e.g. Charlop-Christy *et al.*, 2000; Charlop & Milstein, 1989).

In the Gena *et al.* (1996) study, a combination of reinforcement, modeling, and verbal prompting was used to teach adolescents with autism to display contextually appropriate affect. A multiple-baseline design was used across three or four response categories per youth. Appropriate affective responding included facial and verbal components associated with the display of emotions that were congruent with the context of scenarios presented by a therapist. During treatment, reinforcers – in the form of verbal praise and tokens – were delivered contingent on correct responding during training trials, while reinforcement was withheld during probe trials. Correction procedures were also used during training trials exclusively, and included modeling and verbal prompting. The results indicated a systematic increase in appropriate affective responding for all response categories across participants, on both training and probe trials. Those effects generalized across therapists and settings, and maintained over a period of 1 month.

The *in-vivo* modeling used in the Gena *et al.* (1996) study, though effective in producing imitative affective behavior, was a tiring procedure for therapists to use and difficult to use with consistency – it required extensive training time to ensure consistency across therapists who participated in the study (Gena, 1994). Observational learning, whether achieved through participation of typical peers used as “confederates,” (e.g. Cook & Apolloni, 1976; Odom *et al.*, 1985) or through the use of video modeling (e.g. Charlop-Christy *et al.*, 2000), appears to be a powerful means toward improving the social skills of children with disabilities, in general, and children with autism in particular. The encouraging results of Charlop-Christy and her colleagues, who identified the use of video modeling as most successful in helping children with autism develop social skills, provided an incentive for the researchers of the present study to assess whether it would similarly

facilitate the acquisition of appropriate affective responding.

The present study has multiple purposes. Firstly, to extend the Gena *et al.* (1996) study in three important ways: (a) to younger children – specifically, to preschoolers with autism, since early intervention for children with autism is crucial for their progress in the long run (e.g. Fenske, Zalsenski, Krantz, & McClannahan, 1985); (b) to a more naturalistic setting, the participants’ homes; and (c) to a less structured context – play activities. Another important purpose was to identify procedures that require less effort and training on the part of the therapist, than *in vivo* modeling required in prior research (Gena, 1994), as well as maximize consistency of the modeling procedures. Video modeling was considered to be a teaching strategy of choice, not only because it requires less effort on the part of the therapist and maximizes consistency of treatment, but also because it provides opportunities for observational learning – a teaching strategy proven to be effective in improving a great array of skills in children with autism. Thus, one of the experimental questions set out by the present study was whether observational learning through the use of video modeling, and in conjunction with reinforcement contingencies, would be effective in the acquisition of appropriate affect in preschoolers with autism. A final purpose of the present study was to assess generalization of the treatment outcome across three dimensions: (a) new people, including not only therapists, but most importantly the mothers of the participants, since mothers are usually the primary caretakers of young children and as such crucial for the child’s social development; (b) untrained scenarios; and (c) time, by assessing whether acquired affective responding was maintained during 1-month and 3-month follow ups in the absence of intervention.

METHOD

Participants

Three preschool children with autism, Eleni, a girl living in Athens, Greece, and two boys, Billy and Mike, living in New York, participated in the present study. The names we used for the purposes of this study are pseudonyms. All three children met the criteria for Autistic Disorder upon their evaluation at State hospitals, before their enrollment in home-based programs with a behavior analytic orientation,

as well as by licensed psychologists who worked for the agencies that provided home-programming services and used the diagnostic criteria of DSM-V. The participants were in treatment for an average period of two years prior to their entry in this study. They had an extensive vocabulary, spoke in full sentences, had developed generalized imitation repertoires, could discriminate and name verbal and nonverbal responses of others, but were not familiar with video modeling procedures. All participants were able to label the facial expressions associated with joy, anger, and sadness both from pictures and on live people. Nevertheless, they did not display affective responses according to social demands. In situations that called for sympathy, disapproval or appreciation, their responses were limited to "yes" or "no" statements, or they did not respond at all. All participants were familiar with token reinforcement systems. They attended regular preschool for part of the day, escorted by shadow teachers who were thoroughly trained in behavior analytic intervention. In that context they received social skills training, such as learning to take an initiative to interact with peers, or responding to their peers' invitations for play. At the beginning of the study Eleni was 5 years and 7 months and functioned within the mild range of mental retardation (no standardized tests were administered because such test for intellectual or adaptive skills are not available in the Greek language). Billy was 4 years and 4 months old and had received a standard equivalent score of 99 on the Peabody Picture Vocabulary Test (PPVT; Dunn & Dunn, 1981) and an IQ score of 50 on the revised Stanford-Binet. His IQ score was based on the average of the Verbal Reasoning, Abstract Reasoning, and Short-Term Memory subscales, because he reached no basal on the Quantitative Reasoning subscale. His adaptive behavior composite on the Vineland Scale was 78. Mike was 3 years and 11 months, at the beginning of the study, and had received a standard equivalent score of 113 on the PPVT, and an IQ score of 100 on the revised Stanford-Binet. His IQ score was also based on the average of the same subscales as Billy, because he also reached no basal on the Quantitative Reasoning subscale.

Settings and Therapists

The study took place in the preschoolers' homes. For the third subject, Mike, the setting changed many times throughout the study to different rooms and

different households due to repeated family moves. Favorite toys and edibles were used during sessions as well as a television set and a VCR for the purposes of video modeling. During the video-modeling intervention phase, three separate videotapes were used, one for each affect category (sympathy, appreciation and disapproval). In each tape, a peer displayed an appropriate affective response associated with one of the affect categories. The peer was of the same sex and age as the target child. The therapists who conducted the sessions were three psychologists (one for each child) with both academic and clinical training in behavior analysis and several years of experience in working with children with autism. They were well familiar with the children since they had worked with them from the beginning of their treatment.

Response Definitions

Appropriate affective behavior was defined as a cluster of responses. Specifically, affective responses were scored as correct when: (a) the participant displayed both the operationally defined verbal and facial responses that were congruent with the scenarios presented by the therapist (see Table I), (b) the target responses were emitted within 5 s following the presentation of scenarios, and (c) the child made eye contact with the therapist while emitting appropriate affective responses. Failure to meet any one of those conditions resulted in the child's response being scored as incorrect. The participants were trained to respond with appropriate affect in the following response categories: appreciation, sympathy, and disapproval. The scenarios, materials, and verbal target responses varied across participants according to their preferences, experiences, and verbal skills. Some examples of scenarios and their corresponding target responses are the following: (a) For the response category of showing disapproval, the therapist might say, "I'm going to throw Mr. Count in the garbage" (Mr. Count is a Sesame-Street character that one of the participants liked); or (b) for the response category of showing appreciation the therapist might use a puppet to say, "I'm not going to play 'hide and seek' with you" (for a participant who liked to play "hide and seek"). The first scenario called for an affective response of showing disapproval. Specifically, an appropriate verbal target response would be, "Hey! That's not nice!" or "Don't do that! I don't like that!" Participants were trained to emit at least three different verbal responses in

Table 1. Affective Response Categories and Definitions of Appropriate Affective Behavior

Response categories	Appropriate affective behavior
Showing sympathy	Directing eye gaze toward therapist; providing an appropriate verbal response (e.g., "I'm so sorry for you."); maintaining a serious tone of voice and a serious facial expression.
Showing appreciation	Directing eye gaze toward therapist; providing an appropriate verbal response (e.g., "Thanks. I would really like that."); smiling or laughing at the same time and maintaining a high tone of voice.
Showing disapproval	Directing eye gaze toward therapist; providing an appropriate verbal response (e.g., "Don't do that. That's not nice"); maintaining a serious tone of voice and an upset facial expression (e.g., frowning, wrinkling nose).

each category of affective behavior. Even though verbal responses were tailored to specific scenarios, the same facial expression and gesture were required for all responses within the same response category. For example, for showing appreciation, the appropriate verbal responses were accompanied by smiling or laughing as well as raising the hands to show excitement, whereas, shaking the head accompanied statements of disapproval. In addition, specific tones of voice were targeted for each response category. Responses of appreciation and disapproval called for a higher and more animated tone of voice, whereas a lower and flatter tone was used for sympathy statements. Even though specific gestures were modeled and practiced during training for each affective response category, they were not included in the criteria for reinforcement delivery.

Scenarios and Play Context

Scenarios were questions or statements made by the therapist in the context of pretend play. The context was similar for all participants, but the content of the scenarios varied, as stated above. Pretend play included manipulating toys and various other objects that interested the participants. For example, in a scenario from the response category of showing appreciation the therapist might ask, "Do you want to play with Barbie?" while simultaneously presenting a facial expression of excitement (i.e., smile, eyes open wide) and using a high tone of voice. The content of the scenarios along with the congruent facial expression and tone of voice of the therapist were to function as discriminative stimuli to set the occasion for correct affective responding.

A total of 140 scenarios per child were presented throughout the study, 47 from the response category of showing appreciation, 47 from the category of showing disapproval, and 46 from the category of showing sympathy. Fourteen to 15 scenarios from each response category were used during probe trials, and the remaining 31 to 33 were used during training

trials. Those scenarios were randomly assigned to ten stimulus sets. Each set included 10 training and 4 probe scenarios. Probes were interspersed among training trials using the following randomization rules: (a) during the first and the last trial of each stimulus set, only training scenarios were used and (b) probes were not used during consecutive trials. Those rules have been used in prior research with children with autism to avoid lack of motivation that may result from infrequent opportunities to receive reinforcement (e.g. Gena *et al.*, 1996; Young *et al.*, 1994). Furthermore, stimulus sets were randomly and equally distributed to groups A and B. The distribution of stimulus sets to two groups served the purpose of using new and untrained scenarios during the second phase of the intervention, rather than using the same scenarios that were already trained during the first intervention phase. Thus, Eleni trained with stimulus sets of group A during the first phase of intervention, but during the second phase she trained with the stimulus sets of group B, which were similar but not identical to those of group A. Both groups of sets had been used for the *in-vivo* as well as the video-modeling condition.

Procedure

General Procedure

Experimental sessions were conducted two to four times per week. Each session lasted approximately 15–20 min. and consisted of 14 trials, 10 of which were training trials and four were probe trials. The scenarios used for each session comprised a stimulus set. A total of 20 stimulus sets were used for each child in consecutive order to avoid memorization of specific scenarios, as well as to provide a broad enough sample. The stimulus sets were presented in random order throughout experimental conditions. Stimulus sets from both sets A and B were presented during follow-up sessions, which were conducted in the same manner as baseline sessions.

A session began with the therapist leading the child to the play area, where they sat on the floor facing each other. Several toys and edibles were placed next to the therapist and out of the child's reach, along with a clipboard with a sheet that indicated the scenarios that would be presented during each session. A clipboard with ten stickers (token board) was placed between the child and the therapist and a television set and a VCR were also placed near the therapist to be used for video modeling purposes. The therapist presented each scenario while manipulating the corresponding materials in a playful manner, waited up to 5 s for the participant to respond, and finally delivered the designated consequences. At the beginning of each trial, participants were to establish eye contact and a neutral facial expression before the therapist presented a scenario. All sessions were videotaped for scoring purposes, using a JVC Compact VHS GR-AX68 camera.

Experimental Conditions and Design

A multiple baseline design across subjects with a return to baseline condition was used to assess the effectiveness of *in-vivo* modeling or video modeling, in combination with reinforcement and prompting procedures, in producing contextually appropriate affective responding. After the first baseline condition, the first intervention was introduced; once criterion performance levels were demonstrated (at least 90% correct responding), the second baseline condition was introduced. The second intervention condition started after performance in the second baseline reached low and stable levels of responding. The order of presentation of the two intervention procedures varied across subjects to control for order effects. Specifically, for Eleni and Mike, the *in-vivo*-modeling condition was introduced first using the stimulus sets of Group A, which were also used during Baseline I. The video-modeling procedure was introduced during the second phase of treatment, using the stimulus sets of Group B, which were also used during Baseline I and Baseline II. For Billy, the order in which the two procedures were used was reversed: video modeling was used first and *in-vivo* modeling was introduced next. The groups of stimulus sets were also assigned to the experimental conditions in reverse order.

During baseline, generalization, and follow-up sessions contingent reinforcement for appropriate affective behavior was withheld, but verbal praise was

delivered contingent on the child's attending, appropriate sitting, and compliance. If during those conditions the child emitted an appropriate affective response, the therapist waited 5 s before providing behavior-specific praise for attending and other appropriate behavior.

During training trials, two error-correction procedures were used. For the *in-vivo* modeling condition, the experimenter modeled the appropriate response and guided the participant to match her model, using verbal and gestural prompts, if necessary. For example, if in response to a scenario requiring the participant to show sympathy, the participant said, "I am so sorry for you" in an appropriate tone of voice, but smiled at the same time, the therapist modeled a serious facial response and said, "Look serious like me and say, 'I am so sorry for you.'" The correction procedure was used up to three times per trial. Reinforcement was delivered contingently upon the display of appropriate affective responses and included both verbal praise and token delivery.

During probe trials, neither correction nor reinforcement procedures were used. Instead, the therapist praised the participant for sitting on the floor appropriately, making eye contact, and for other appropriate responses, independent of his or her affective behavior. If a participant emitted an appropriate affective response, congruent with the scenario presented by the therapist, verbal praise was withheld for at least 5 s. Experimental sessions ended with the participant exchanging his or her accumulated tokens for preferred activities, objects, or snacks.

For the video-modeling condition, the same procedures were used as in the *in-vivo*-modeling condition, with the exception of the error correction procedure. Instead of modeling the appropriate response, the therapist played the videotaped segment of a peer (each of whom was selected to match the gender and the age of the participant for whom he or she modeled) modeling the correct response, and said to the participant, "Look at (name of peer). You do it too." The peers modeled an appropriate verbal response in conjunction with an appropriate facial expression for each category of affective responding. The video model was presented up to three times and if the participant did not match the model, the experimenter said, "No, this is what (name of peer) did," providing a verbal and facial display of the correct affective behavior, and moved on to the next trial.

Generalization Across People

A new therapist and the child's mother tested for generalization across people. The procedures used during generalization sessions were the same as those used in baseline conditions. The new therapist and the child's mother conducted three generalization sessions during the initial baseline condition and three additional sessions following the final treatment session.

Follow-Up Sessions

The primary therapist conducted 1-month and 3-month follow-up sessions using the same procedures as those used during baseline. Between the last treatment and the first follow-up session, the participants received no training on affective responding.

Measures of the Independent Variable

Data were collected on all measures of the independent variable including presentation of scenarios, reinforcement contingencies, and error-correction procedures across all sessions of all experimental conditions. The data for the independent measures, as with the dependent measures, were coded from videotapes. During both training and probe trials and throughout the study operations of the independent variable were established as programmed with 1% error in the delivery of reinforcement and no errors in the application of the error correction procedures and in the presentation of scenarios.

Interobserver Agreement

The primary therapists were two psychologists working with children with autism, who also served as the primary data collectors. In addition, two graduate students with training in behavior analytic intervention conducted the generalization sessions, who along with an undergraduate psychology student, collected data for interobserver-agreement purposes. Interobserver agreement was calculated by using a point-by-point comparison method (Kazdin, 1982), on 30–50% of the sessions for all experimental conditions, including generalization and follow-up. Interobserver agreement on the dependent measures, across all experimental conditions, ranged from 90 to 100% on both probe and training trials. The observers were trained to criterion of 90% agreement, with the primary therapists, before starting independent recording. Interobserver agreement data were also

collected for measurements of the independent variables and yielded 100% agreement invariably.

RESULTS

Figure 1 shows the percentage of appropriate affective responding for all three children during training and probe trials and for all phases of the experiment.

The closed circles in the figure represent percentages of appropriate affective responses emitted during training trials, and the open circles represent performances during probe trials.

During baseline I the three participants demonstrated very low levels of appropriate affective responding, which increased dramatically with the introduction of the intervention procedures on both training and probe trials. During the baseline II condition treatment outcomes were reversed, but were reinstated during the second intervention phase. Reversal and reinstatement of appropriate affective responding occurred equally on training and probe trials for all three children. Specifically, Eleni's performance during baseline I was invariably at zero. She emitted no correct responses in any of the three response categories. With the introduction of *in-vivo* modeling and the other treatment contingencies, her affective responding increased to 50% within three sessions on training trials and within four sessions on probe trials. It continued to increase, reaching 90% on training trials and 100% on probe trials within eleven sessions. Withdrawal of treatment during baseline II resulted in a steady decrease in the percentage of appropriate affective responding. When affective responding stabilized to a low level (0 and 25%), the second intervention phase was introduced. During the video-modeling condition, appropriate affective responding increased to 60% in the first session on training trials and to 50% on probe trials. The percentage of appropriate affective responses continued to increase, reaching 100% on both training and probe trials by the tenth session. Treatment outcomes generalized to both a new therapist and to the participant's mother. Further, during the 1-month follow-up session, trained affective responses were emitted 100% of the time, whereas probed responses were maintained at 70%. During the 3-month follow-up session, Eleni's appropriate affective responding maintained at 80% during training trials and 100% during probe trials.

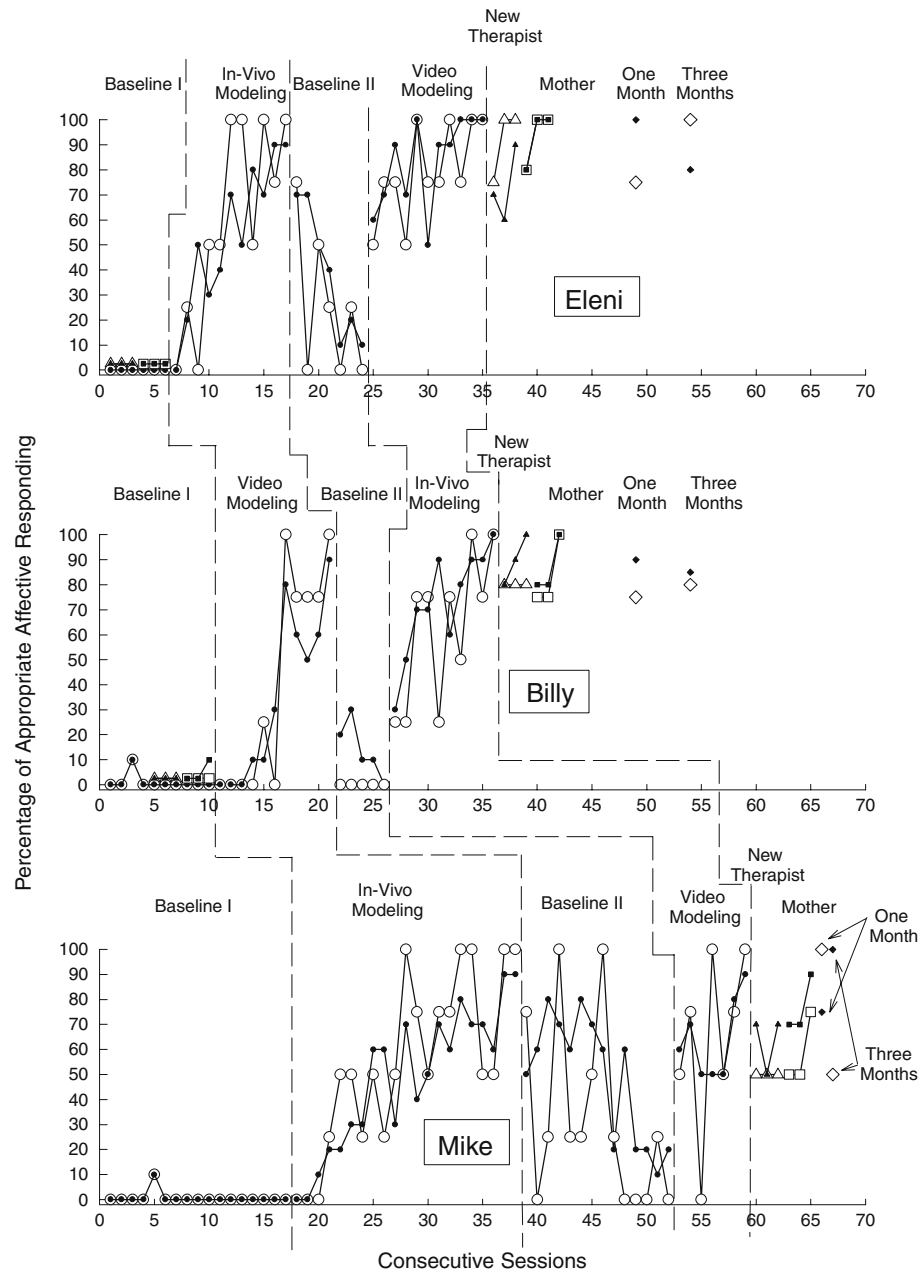


Fig. 1. Percentage of appropriate affective responding of three children with autism, across three response categories, over consecutive sessions. The closed symbols depict training-trial data and the open circles depict probe-trial data.

The percentage of Billy's appropriate affective responses during the baseline I condition was near zero. During the video-modeling condition, his affective responding increased to 80% on training trials and 100% on probe trials within six sessions, and after a slight decrease for three consecutive sessions, it increased again to 90% on training and 100% on

probe trials. During baseline II, affective responding decreased to 0% for both training and probe trials within five sessions. Appropriate affective responding increased steadily during the *in-vivo* modeling condition, at the end of which it reached 100% on both training and probe trials. Billy's affective responses ranged from 75 to 100% during generalization

sessions with the new therapist and his mother. He also maintained appropriate affective responding at an average of 80% on training and probe trials during the 1-month and 3-month follow up sessions.

The third participant, Mike, remained in baseline I for 17 sessions, during which he emitted only two appropriate affective responses. With the introduction of *in-vivo* modeling and the other treatment contingencies, his data showed an ascending trend with some variability. Appropriate affective responding reached 100% on training trials and 90% on probe trials within 20 sessions. During the baseline II condition, his data reversed progressively and over the course of 13 sessions. Finally, during the video-modeling condition, Mike showed rapid acquisition and reached 90–100% appropriate affective responding within seven sessions. Mike learned the appropriate verbal responses rapidly, but had difficulty displaying the requested facial expressions. During generalization sessions with the new therapist and Mike's mother, his appropriate affective responding ranged from 50 to 75% and maintained at high levels during the 1-month and 3-month follow-up sessions. Nevertheless, from anecdotal observation it was noted that the quality of his facial expressions was not preserved as well as his verbal responses three months after treatment.

Acquisition of appropriate affective responding during the second intervention phase was achieved in fewer sessions than in the first intervention phase, for all three participants, regardless of whether *in-vivo* or video modeling was used.

DISCUSSION

This study demonstrated that contextually appropriate affective responding can be taught to preschool children with autism at home and in the context of play activities using *in-vivo* modeling or video modeling, along with reinforcement and prompting procedures. In addition, those responses generalized to new, untrained scenarios, to new therapists, and to the child's mother, and maintained during 1- and 3-month follow-up sessions.

The findings of the present study replicate the Gena *et al.* (1996) results, pertaining to the acquisition and generalization of appropriate affect by youths with autism. The findings of both studies provide support for the hypothesis that the affective behavior of children with autism can be treated as operant behavior. It is interesting that the

participants spoke in full sentences and had extensive training in motor and vocal imitation before they entered the study, yet did not begin to imitate the therapist's models of affective responding until reinforcement and prompting procedures were implemented. The importance of environmental contingencies in shaping affective responding has also been emphasized in cross-cultural research, which demonstrates that affective responding is, up to a certain degree, culture-specific (Davitz, 1964; Ekman, 1984). The present study, though conducted in two countries, did not specifically examine differences in affective responding between the two cultures. Nevertheless, replication of the treatment outcome in both countries offers support to the generality of the treatment outcome.

The therapists' facial displays, accompanied by relevant verbal statements, apparently functioned as discriminative stimuli for appropriate affective responding. The present study, however, did not address whether it was the nonverbal aspects of the communication (tone of voice, gestures, and facial displays), the content of the scenarios, or the combination of both that acquired discriminative properties. Prior research indicates that individuals with autism are impaired in understanding the emotional expression of others as well as the nonverbal elements required to comprehend social exchanges (Philippot, Feldman, & McGee, 1992). Two important implications are derived from our study: children with autism may both learn to discriminate nonverbal elements of communication and also adjust their affective behavior called for in various social situations. Failing to comprehend the complex discriminative stimuli associated with social events and consequently adjusting one's affective behavior accordingly, may impede the acquisition of the pragmatics of language, which ultimately may lead to poor social interactions. Thus, the clinical implications of the present study are important for children with autism who have difficulties comprehending the pragmatics of language and developing appropriate social repertoires.

Another important finding replicated by the present study, though not presented in the results, is the rate of acquisition of the nonverbal as opposed to the verbal aspects of the target affective responses. The participants invariably required more time and practice to acquire the facial displays than the verbal responses associated with affect. Mike, for example, had great difficulty imitating facial movements, such as frowning, yet had no difficulty recognizing the

therapist's facial cues, to the point that sometimes he would begin to respond appropriately before the therapist had a chance to complete a scenario. Thus, analyzing the learning process in the modification of affective behavior of individuals with autism, we may conclude from both the Gena *et al.* (1996) and the present study that adjusting one's facial expression to match social demands is more of a challenge for individuals with autism than is comprehending the verbal and nonverbal aspects of social exchanges that require some type of an affective response.

Considering the importance of identifying procedures that could help people with autism overcome deficits in the area of affective behavior, we need to emphasize that the present study provides two effective ways to ameliorate such problems. The *in-vivo* as well as the video modeling procedures, in conjunction with reinforcement contingencies, were both effective in teaching appropriate affective responding to the preschoolers with autism who participated in the present study. Provided the small number of participants we may not conclude whether one procedure would result in faster acquisition over the other. The effect of a faster acquisition during the second intervention phase can be attributed to multiple – treatment interference. According to Kazdin (1982), the use of a baseline phase between interventions does not necessarily eliminate the possibility that behavior may be influenced by the mere fact that another intervention preceded it. Even though the data of the present study do not reflect differences between the two intervention procedures, from anecdotal observations, it appeared that acquisition of appropriate tone of voice and facial expressions were more expedient when video modeling was used. It would be interesting for future research to assess whether learning to imitate facial expressions through video modeling prior to the intervention could facilitate the acquisition of contextually appropriate affective responding.

The present study addressed the generalization of affective behavior across people and scenarios, and over a time span of 3 months. The appropriate affective responding of all three children generalized to their mothers, to new therapists, and to untrained scenarios, and maintained for 3 months following termination of treatment. An experimental analysis of the factors that promoted generalization was not conducted, yet, we may hypothesize that generalization effects may be attributed to various factors. First of all, the diversity of the scenarios used throughout the study, which exemplifies the condition of

“training sufficient exemplars” according to Stokes and Baer (1977), or varied training exemplars (Brown & Odom, 1994; Jahr, Eldevik, & Eikeseth, 2000; Kazdin, 1994) is a factor that could have promoted generalization across stimuli. The present study is one of the few that used this generalization strategy to promote transfer of complex social skills under novel stimulus conditions, but it has also been used successfully for other types of social behavior, such as cooperative play (Jahr *et al.*, 2000) and conversational interactions (Hughes, Harmer, Killian, & Niarhos, 1995). Secondly, the combination of reinforcing a variety of correct responses and providing a play context, in which training trials were impeded in the present study, provided a case of training multiple stimulus and response exemplars (Stokes & Osnes, 1986) or “training loosely,” which may promote generalization as well (Stokes & Baer, 1977). Finally, the generalization and maintenance of the three preschoolers' appropriate affective responding may be attributed to the fact that during training, the combination of procedures used were the ones identified in the literature as most efficient in promoting generalization and maintenance (e.g., Chandler, Lubeck, & Fowler, 1992; Elliott & Gresham, 1993). Namely, that combination included both antecedent and consequent strategies in the form of prompting and positive reinforcement respectfully, which may strengthen acquired responses more so than experimental procedures that employ either antecedent or consequence manipulations alone. In addition to those manipulations, two of the procedures referred to as “generalization facilitators” (Elliott & Gresham, 1993; Stokes & Osnes, 1989) were employed in the present study: (a) teaching responses that are likely to be maintained by naturally occurring contingencies (i.e. appropriate affect) and (b) training across stimuli common to the natural environment (i.e. having the children's therapists and mothers participate in the study). Those procedures may function in the natural environment as mediators of the treatment outcome.

Identifying procedures that help children with autism acquire and generalize appropriate affective responding and that can be used at home allows for the use of such procedures with very young children. Such early intervention may promote social interactions at a very young age and prevent the social isolation that often stigmatizes children with autism. A limitation of the present study was that generalization was not assessed across environments or contexts outside of the home, such as a playground or the

children's preschool and during the children's naturally occurring interchanges. It would be very interesting for future research to assess whether generalization can be obtained in school settings and with familiar people in the course of everyday activities, as well as whether such training can facilitate the acquisition of novel affective responses at home and in the school. Another empirical question is whether parents or siblings could teach appropriate affective responding to children with autism as a means of improving their communication with family members.

The Gena *et al.* (1996) and the present study are not but the first steps, within the behavior analytic framework, toward helping children with autism to develop appropriate affective repertoires. They do not address more complex issues that researchers with a developmental orientation have pointed out, such as identifying the skills and the procedures that would help children with autism become "intentional, interactive individuals" (Greenspan & Wieder, 1998; Prizant & Rubin, 1999). When compared with typical children and children with other developmental disabilities, children with autism have fewer interactions with their caregivers (Sigman, Mundy, Sherman, & Ungerer, 1986), make less eye contact, may avoid physical contact, have difficulties discriminating the affective expressions of others, and they often fail to reciprocate affective expressions (Schreibman & Charlop-Christy, 1998; Snow, Hertiz, & Shapiro, 1987; Weeks & Hobson, 1987). Establishing shared attention, verbal and non-verbal reciprocal interchanges, communicative intent, and other important social-communicative responses that children with autism could use in a generalized manner, would doubtlessly change significantly their ability to socialize. One of the research questions that could address those issues would be to examine the relation between establishing a theory of mind and developing appropriate social-affective repertoires. If the development of appropriate affect is closely affiliated or dependent upon understanding other people's thoughts and intentions, then acquiring such an understanding should favor the development of socially appropriate affective expressions as well as the ability to initiate social interactions. As recent empirical studies investigating the correlation of affective behavior and overall social understanding and social competence demonstrate, we are still far from having a clear account of the social deficits associated with autism (e.g., Travis, Sigman, & Ruskin, 2001). Similar questions regarding the achievement of widespread improvements in the domain of social behavior and

self-initiated learning have also been posed within the behavior analytic framework, seeking to identify the types of responses that would promote such improvements. Specifically, Robert Koegel and his colleagues have identified "pivotal" behaviors, such as self-management and self-initiated learning, the acquisition of which may have widespread impacts on the overall functioning of children with autism (e.g., Koegel, R., Koegel, L., & Carter, 1999; Koegel, L., Koegel, R., Harrower, & Carter, 1999). It would be very interesting for future research addressing the affective behavior of children with autism to investigate whether acquisition of "pivotal" skills or skills associated with developing a theory of mind would automatically produce wide-spread improvements in the affect of children with autism.

The area of affective responding of people with autism is certainly under-researched, yet the expression and comprehension of emotion are crucial elements for a person's integration in society. Planning to teach individuals with autism contextually appropriate affective behavior is an important element of their education that could start early on in development and minimize the stigma associated with their social behavior.

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